

Optimization: Critical Points & Lagrange Multipliers



Finding critical points

- 1 Find the critical point(s) of $f(x, y) = x^2 + 3y^2 - 4x + 6y$.
 - 2 Find the critical point(s) of $f(x, y) = xy - x^2 - y^2$.
 - 3 Find the critical point(s) of $f(x, y) = x^3 - 3xy + y^3$.
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The Hessian and the Second Derivative Test

- 4 Classify the critical point $(2, -1)$ of $f(x, y) = x^2 + 3y^2 - 4x + 6y$ using $D = f_{xx}f_{yy} - (f_{xy})^2$.
 - 5 Classify the critical point $(0, 0)$ of $f(x, y) = xy - x^2 - y^2$.
 - 6 Classify the critical point of $f(x, y) = x^2 - y^2$.
 - 7 Classify both critical points $(0, 0)$ and $(1, 1)$ of $f(x, y) = x^3 - 3xy + y^3$ (found above).
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When the Second Derivative Test fails

- 8 Show that the Second Derivative Test is inconclusive at the critical point $(0, 0)$ of $f(x, y) = x^4 + y^4$.
 - 9 Determine directly (without the Second Derivative Test) whether $(0, 0)$ is a local min, max, or saddle for $f(x, y) = x^4 + y^4$.
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Lagrange multipliers: setting up the system

- 10 Set up the Lagrange multiplier equations to optimize $f(x, y) = xy$ subject to $g(x, y) = x + y - 10 = 0$.
 - 11 Set up the Lagrange multiplier equations to optimize $f(x, y) = x^2 + y^2$ subject to $g(x, y) = xy - 1 = 0$.
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Solving Lagrange multiplier problems

- 12 Maximize $f(x, y) = xy$ subject to $x + y = 10$.
- 13 Minimize $f(x, y) = x^2 + y^2$ subject to $x + y = 4$.

- 14 Solve the system from the earlier setup question to minimize $f(x, y) = x^2 + y^2$ subject to $xy = 1$.
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A geometric Lagrange multiplier application

- 15 Find the point on the line $x + y = 6$ closest to the origin, using Lagrange multipliers on $f(x, y) = x^2 + y^2$.
- 16 A rectangle has perimeter 20. Use Lagrange multipliers to maximize its area $A = xy$ subject to the constraint $2x + 2y = 20$ (i.e., $x + y = 10$).
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Lagrange multipliers with three variables

- 17 Set up (but do not fully solve) the Lagrange multiplier equations to minimize $f(x, y, z) = x^2 + y^2 + z^2$ subject to $g(x, y, z) = x + y + z - 6 = 0$.
- 18 Solve the three-variable problem set up above.
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Interpreting the meaning of λ

- 19 In the rectangle-area problem above ($x = y = 5$, $\lambda = 5$), briefly explain what λ represents in economic/physical terms: the approximate change in the optimal value of f per unit increase in the constraint constant.
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Optimization on a closed region (global extrema)

- 20 Explain the general procedure for finding the global maximum and minimum of a continuous $f(x, y)$ on a closed, bounded region R .
- 21 Find the only interior critical point of $f(x, y) = x^2 + y^2 - 2x$ on the closed disk $x^2 + y^2 \leq 4$, and evaluate f there.
- 22 For the same $f(x, y) = x^2 + y^2 - 2x$ on $x^2 + y^2 \leq 4$, find the maximum of f on the boundary circle $x^2 + y^2 = 4$ using Lagrange multipliers.
- 23 Combine the results above to state the global maximum and minimum of $f(x, y) = x^2 + y^2 - 2x$ on the disk $x^2 + y^2 \leq 4$.
- 24 Explain why the Extreme Value Theorem guarantees this problem has a global max and min, and why that guarantee would fail on the OPEN disk $x^2 + y^2 < 4$.

- 25** Find the global maximum and minimum of $f(x, y) = xy$ on the closed square $-1 \leq x \leq 1$, $-1 \leq y \leq 1$, by checking the interior critical point and all four corners (the boundary edges of a square are easiest checked at their endpoints for a bilinear function like xy).